Mrinaal Dogra

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Professional Summary

Senior Machine Learning Engineer with 5+ years of experience specializing in Agentic AI, Large Language Models (LLMs), and On-Device ML. Proven track record at Samsung R&D leading full-cycle engineering from data processing to edge deployment. Currently advancing expertise in Generative AI at UCSD, seeking to leverage deep architectural knowledge of distributed systems, AWS/MLOps, and model optimization to build scalable AI solutions.

Technical Skills

Languages: Python, C/C++, Java, Kotlin, Golang, SQL, Bash
Frameworks & Libraries: PyTorch, TensorFlow, LangChain, Transformers, Scikit-learn, Pandas, NumPy, DL4J
AI Concepts: GenAI, LLMs, RAG, Agents, Deep Learning, Federated Learning, Edge ML, NLP, CV, MCP
Cloud & Tools: AWS (EC2, Lambda, S3), Docker, Git/GitHub, Linux, FastAPI

Professional Experience

Kognitos — *San Jose, CA*
Machine Learning Engineer Intern — *Jun. 2025 - Sep. 2025*

- Architected a greenfield **Agentic AI pipeline to automate Standard Operating Procedure (SOP) generation** from desktop recordings, introducing a novel, **multi-modal** capability to the product suite where no automation previously existed.
- Engineered a full-stack solution integrating **Computer Vision (key-frame extraction), OCR, and LLM orchestration**; packaged the system as a **containerized web application** deployed on **AWS EC2**, enabling immediate internal testing across diverse inputs.
- Conducted rigorous benchmarking and prompt engineering across GPT and Gemini series, systematically evaluating trade-offs between reasoning accuracy, structured output quality, token costs, and inference latency to define the production rollout strategy.
- Developed a **scalable PII masking microservice** using the GLINER model; evolved the architecture from a single-instance EC2 REST API to a **serverless, Dockerized AWS Lambda** function with role-based access control to ensure seamless product integration.

Samsung R&D Institute India - Bangalore (SRI-B) — *Bengaluru, India*
Lead Engineer, Machine Learning — *Mar. 2023 - Aug. 2024*

- Spearheaded the **end-to-end engineering of four edge-based LLM personalization features** for Android, translating high-level product goals into efficient on-device architectures designed to generate contextual insights and improve feature discoverability.
- Fine-tuned the **FLAN-T5** model for mobile deployment using **domain-specific prompt tuning**, achieving **~92% intent prediction accuracy** by engineering a hybrid *Semantic + Exact Match* evaluation metric to overcome significant data labeling noise.

Senior Software Engineer, Machine Learning — *Mar. 2021 - Feb. 2023*

- Engineered an **end-to-end On-Device ML system** for user **boredom detection**, managing full lifecycle from raw data processing to building an Android app that demonstrated real-time inference with **~50ms latency** to prove technical feasibility on edge hardware.
- Designed a **privacy-preserving Federated Learning (FL)** architecture for demographic prediction, executing extensive experiments across 20+ model configurations and aggregation algorithms to benchmark privacy-utility trade-offs against centralized training baselines.

Software Engineer, Machine Learning — *Jun. 2019 - Feb. 2021*

- Engineered a **privacy-preserving Next App Recommendation** model (published in **IEEE ICSC 2022**) achieving a **99% size reduction** over existing SOTA, specifically designed to minimize communication overhead and bandwidth costs for FL updates.
- Implemented the complete on-device training and inference pipeline using Java and Deeplearning4j (DL4J), deploying the solution to a **500+ device User Trial (UT)** network to validate distributed learning stability and inference performance in real-world conditions.

Applied ML Projects & Research

SimWorld (NeurIPS 2025) | *Research Project, UCSD* — *Dec. 2024 - May. 2025*

- Integrated interactive environmental assets and functional logic into Unreal Engine 5, establishing a robust communication layer that enables precise programmatic control of physical world states directly from the core simulation engine.
- Enhanced the simulation's fidelity by implementing dynamic object behaviors and physics-based interactions, thereby creating a rich, reactive testing ground for training embodied agents in complex, open-ended physical environments.

Multi-task Learning with ToolkenGPT Framework | *Course Project, UCSD* — *Sep. 2024 - Dec. 2024*

- Adapted the ToolkenGPT framework for Llama-3.2 (1B), engineering a **novel multi-task learning pipeline** that consolidates toolken fine-tuning across multiple domains into a single training run to explicitly eliminate redundant fine-tuning cycles.
- Validated that the unified multi-task approach maintained performance parity with individual single-task baselines, effectively demonstrating a scalable, compute-efficient paradigm for tool learning that minimizes total GPU resource usage.

Education

M.S. in Computer Science, **University of California, San Diego** (*GPA: 3.96/4.0*) — Sep. 2024 - Mar. 2026
B.Tech in C.S.E., **Indian Institute of Technology, Kanpur (IIT Kanpur)** (*GPA: 9.0/10.0*) — Jul. 2015 - Jun. 2019

Awards and Patents

Patent (US 18/191403): Methods and Electronic Devices for Behavior Detection using Federated Learning
Samsung Excellence Award (Star of the Quarter, 2024): Recognized as a top engineer for outstanding quarterly project impact.
Key Talent Recognition Program Award (2023): Selected among top 10% globally for high-impact contributions & teamwork.